ISPRS Journal of Photogrammetry and Remote Sensing 214 (2024) 21–32

Contents lists available at ScienceDirect

ISPRS Journal of Photogrammetry and Remote Sensing

journal homepage: www.elsevier.com/locate/isprsjprs

CloudSeg: A multi-modal learning framework for robust land cover mapping under cloudy conditions

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ARTICLE INFO

Keywords:
Land cover mapping
Cloud removal
Optical-SAR fusion
Multi-task learning
Knowledge distillation

ABSTRACT

Cloud coverage poses a significant challenge to optical image interpretation, degrading ground information on Earth's surface. Synthetic aperture radar (SAR), with its ability to penetrate clouds, provides supplementary information to optical data. However, existing optical-SAR fusion methods predominantly focus on cloud-free scenarios, neglecting the practical challenge of semantic segmentation under cloudy conditions. To tackle this issue, we propose CloudSeg, a novel framework tailored for land cover mapping in the presence of clouds. It addresses the challenges posed by cloud cover from two aspects: reducing semantic ambiguity in areas of the cloudy image that are obscured by clouds and enhancing effective information in the unobstructed portions. Specifically, CloudSeg employs a multi-task learning strategy to simultaneously handle low-level visual task and high-level semantic understanding task, mitigating the semantic ambiguity caused by cloud cover by acquiring discriminative features through an auxiliary cloud removal task. Additionally, CloudSeg incorporates a knowledge distillation strategy, which utilizes the knowledge learned by the teacher network under cloud-free conditions to guide the student network to overcome the interference of cloud-covered areas, enhancing the valuable information from the unobstructed parts of cloud-covered images. Extensive experiments conducted on two datasets, *M3M-CR* and *WHU-OPT-SAR*, demonstrate the effectiveness and superiority of the proposed CloudSeg method for land cover mapping under cloudy conditions. Specifically, CloudSeg outperforms the state-of-the-art competitors by 3.16% in terms of mIoU on *M3M-CR* and by 5.56% on *WHU-OPT-SAR*, highlighting its substantial advantages for analyzing regions frequently obscured by clouds. Codes are available at https://github.com/xufangchn/CloudSeg.

1. Introduction

Optical remote sensing satellites inevitably encounter the influence of atmospheric clouds when collecting surface information (King et al., 2013; Wang et al., 2021a). This leads to acquired data that may not meet the quality and quantity requirements for remote sensing research and applications, particularly in quantitative remote sensing and time-series analysis. For instance, in regions like Bangladesh and the Amazon Basin where cloud cover is persistent throughout the year (Mitchard et al., 2012; Martins et al., 2018; Tsai et al., 2018; Ahmad et al., 2019), acquiring cloud-free optical images for accurate land cover mapping poses significant challenges. Moreover, for applications requiring timely information, such as agricultural monitoring and disaster emergency management, the potential delay in obtaining cloud-free images could hinder an effective response to the rapidly evolving conditions on the ground. Synthetic aperture radar (SAR), which can observe ground objects regardless of cloudy conditions by using longer wavelengths of the electromagnetic spectrum, has emerged as a valuable complement to optical remote sensing. On the one hand, it can compensate for the missing information in regions affected by clouds, where optical images are unable to provide clear observations. On the other hand, it can provide unique insights that are not available in optical images. Consequently, the fusion of optical and SAR images can provide a more comprehensive and robust representation of the scene, thus essentially overcoming the challenges associated with scene interpretation under cloudy conditions.

Optical-SAR data fusion has been extensively studied and made great progress in the past few years (Schmitt et al., 2017; Prabhakar

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https://doi.org/10.1016/j.isprsjprs.2024.06.001
Received 19 January 2024; Received in revised form 2 June 2024; Accepted 3 June 2024
Available online 10 June 2024

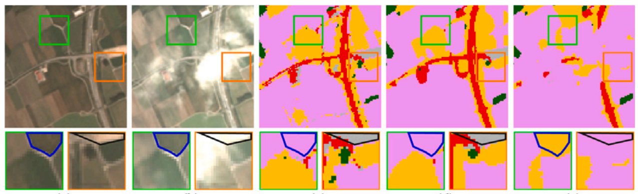

Fig. 1. Analysis of the impact of cloud cover on land cover classification (the details of the local areas highlighted by the green and orange squares are displayed below for better viewing): the influence of clouds on land cover classification extends beyond the areas directly covered by clouds, *e.g.*, the area framed in **black**, also affecting the surrounding cloud-free regions, *e.g.*, the area framed in **blue**. (a) Cloud-free optical image. (b) Cloud-covered optical image. (c) Land cover annotation. (d) Results with the cloud-free image as input. (e) Results with the cloud-covered image as input. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

et al., 2022; Kang et al., 2022). However, the majority of previous methods primarily focus on the fusion of SAR images and cloud-free optical images. There are fewer works considering the practical problem: multi-modal fusion for semantic segmentation of land cover under cloudy conditions, in which the negative impact induced by clouds has to be taken into account. As shown in Fig. 1, models trained on cloud-free images may fail when processing cloud-covered images. What is more, the influence of clouds on land cover classification extends beyond the areas they directly cover. Their presence can even affect the classification results in nearby cloud-free areas. Due to the semantic interdependencies in spatial dimensions, the features of cloudy and cloud-free regions interact with each other and collectively determine the overall land cover classification results. This also implies that considering the features of an individual region is insufficient, and it is necessary to consider the feature interactions across the entire scene. Therefore, in this study, to improve the accuracy of land cover classification under cloud-covered conditions, we primarily focus on the following two perspectives:

- **Reducing Semantic Ambiguity in Cloudy Areas**. In regions covered by clouds, the information of optical images is often distorted or lost, leading to semantic ambiguity in these areas. This ambiguity gives rise to two main issues. On one hand, it directly interferes with the feature fusion process within cloud-covered areas. On the other hand, this interference may further affect the feature representation in nearby cloud-free areas. Therefore, it is essential to ensure accurate identification and utilization of SAR features within cloud-covered regions to reduce semantic ambiguity in these areas and prevent them from misleading adjacent cloud-free regions.
- **Enhancing Effective Information in Cloud-Free Areas**. In cloud-free regions, both optical and SAR imagery present clear and complete land feature characteristics, which are of great value for land cover classification. To improve the model's predictive performance, it is crucial to effectively utilize these feature resources. By delving into the land features within cloud-free regions, it is possible to infer similar or neighboring land features that may exist in cloud-covered areas. However, it is also important to consider potential interference from cloud-covered areas in cloud-free regions. Therefore, corresponding strategies should be adopted to minimize such interference and ensure the accuracy of predictions.

In this work, we propose a novel framework called CloudSeg, which integrates these two perspectives to address the negative impacts of cloud cover. Specifically, it employs a multi-task learning strategy to simultaneously tackle low-level visual task and high-level semantic understanding task, facilitating the collaboration between cloud removal and land cover classification tasks. The learning of the cloud removal task assists the model in acquiring more discriminative features from cloud-covered areas, and effectively mitigating the issue of semantic ambiguity caused by cloud cover. The land cover classification task leverages feature information obtained from the auxiliary cloud removal task, alleviating performance degradation due to information loss of optical images. Additionally, CloudSeg employs a knowledge distillation strategy, where the student network emulates the teacher network's prediction under cloud-free conditions. In this process, the teacher network combines optical and SAR images in cloud-free regions to perform land cover classification, guiding the student network to overcome interference in cloud-covered areas and accurately obtain complete and robust features in cloud-free regions. Extensive experiments on *M3M-CR* and *WHU-OPT-SAR* datasets are conducted to validate the effectiveness of the proposed method. Our CloudSeg remarkably boosts the land cover classification performance under cloud-covered conditions, surpassing state-of-the-art multi-modal semantic segmentation methods by 3.16% in terms of mIoU on *M3M-CR* and by 5.56% on *WHU-OPT-SAR*.

This paper is an extended version of our prior work (Xu et al., 2023), with several improvements: (1) We enhance the CloudSeg architecture by implementing a three-branch network structure. This design is tailored for both modality-specific and modality-shared representation learning, significantly improving the collaboration between cloud removal and land cover classification tasks. (2) We introduce a novel knowledge distillation strategy that aims at bolstering the information retained in the unobstructed portion of optical images, further mitigating the performance degradation caused by cloud cover. (3) We extend our evaluation to include comparisons of state-of-the-art methods, ablation studies, and validation on a new dataset *WHU-OPT-SAR*, providing a more comprehensive and in-depth analysis of the CloudSeg method.

2. Related work

Land cover classification, a long-standing research problem, aims to assign a specific land cover label to each pixel within remote sensing images (Zhang et al., 2019; Helber et al., 2019; Chen et al., 2023). Early approaches to tackle this problem often rely on traditional machine learning classifiers in conjunction with the prior knowledge or hand-designed features, and thereby cannot capture complex high-level semantic information (Huang et al., 2002; Rodriguez-Galiano et al., 2012; Talukdar et al., 2020). In contrast, deep learning methods advance the field by automatically extracting more intricate features, achieving impressive results (Tong et al., 2020; Wambugu et al., 2021). However, the use of fully connected layers as classifiers in deep learning methods leads to the loss of spatial information, resulting in inaccurate identification of land cover boundaries. The recent progress in land cover classification is significantly attributed to representative semantic segmentation networks like FCN (Long et al., 2015), U-Net (Ronneberger et al., 2015), and SegNet (Badrinarayanan et al., 2017), which, by forgoing fully connected layers, achieve significant breakthroughs in the field (Wang et al., 2021b; Tong et al., 2023; Dong et al., 2023). Yet, the reliance on single-modality data frequently proves insufficient in providing comprehensive information, which remains a hurdle in further improving classification accuracy.

To achieve more accurate and comprehensive land cover information, it is essential to explore the use of multi-modal remote sensing data (Hong et al., 2021; Yan et al., 2023). For example, Zhang et al. (2020) employ a multi-scale analysis method (Chu et al., 2020) to fuse features from optical and SAR images, followed by using an SVM classifier for land cover classification in areas with high cloud coverage. Ling et al. (2021) use a stacking approach to merge optical and polarimetric SAR features, applying SVM, RF, and the GoogLeNet model for urban land cover classification in cloud-prone areas. This study also investigates the quantitative relationship between cloud cover and classification accuracy, as well as the impact of SAR on areas

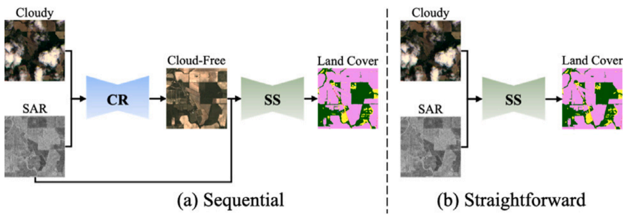

Fig. 2. Two naive strategies to mitigate the interference of cloud cover on land cover classification. (a) The sequential paradigm. (b) The straightforward paradigm. CR is short for cloud removal and SS is short for semantic segmentation.

with and without cloud cover. However, these algorithms treat feature extraction and classifier learning as independent modules, which can lead to sub-optimal classification results.

Currently, many approaches inspired by single-modal semantic segmentation algorithms are proving effective in multi-modal land cover classification (Peng et al., 2019; Liu et al., 2021). For instance, V-FuseNet (Audebert et al., 2018), based on the SegNet model, integrates features from optical and LiDAR data using a residual correction strategy for prediction of land cover maps. Li et al. (2022) propose a multi-modal cross-attention network built upon DeepLabv3+ (Chen et al., 2018), fusing optical and SAR images for land cover semantic segmentation. However, these methods are primarily developed based on high-quality optical images, often assuming consistency or predetermined correlations between different modalities. This assumption becomes problematic when cloud cover obscures parts of the optical images, leading to incomplete or distorted information and mismatches with data from other modalities. This situation complicates the process of multi-modal fusion, potentially reducing the generalizability of the model, resulting in a decline in performance, and occasionally even falling behind the performance achievable through a single modality approach (Xu et al., 2024). To unlock the full potential of multi-modal land cover classification, we propose a novel multi-modal learning framework, CloudSeg, which reduces semantic ambiguity in cloudy areas and enhances effective information in Cloud-free areas to address the specific challenges of semantic segmentation under cloudy conditions.

3. Method

3.1. Naive strategies

To mitigate the interference of cloud cover on land cover classification, one intuitive alternative approach is to adopt a **sequential** paradigm, treating cloud removal as a preprocessing step for subsequent remote sensing image scene understanding, as illustrated in Fig. 2(a). In this paradigm, optical images obscured by clouds are restored using auxiliary information from SAR images. The restored images are then fused with SAR images to predict land cover categories. Ideally, optical images that have undergone cloud removal tend to exhibit clearer features, resulting in more intuitive and accurate land cover classification. However, cloud removal and land cover classification, are optimized independently (Sarukkai et al., 2020). Most of the cloud removal methods rely on per-pixel loss functions (Meraner et al., 2020; Wen et al., 2021; Xu et al., 2022; Wen et al., 2021), which enable better visually good-looking removal of the clouds. However, the per-pixel loss functions ignore semantic information. The image details along with the corresponding discriminative feature may inevitably be damaged. Therefore, cloud removal as a preprocessing step may not achieve optimal results in the downstream land cover classification task.

Another potential approach involves using the original optical images with cloud cover for land cover classification, *i.e.*, the **straightforward** paradigm, as shown in Fig. 2(b). In this strategy, cloudy optical images are directly fused with SAR images to predict land cover types. The core idea of this approach is to include cloud-covered observations in the training dataset, allowing neural networks to become familiar with the presence of clouds (Gawlikowski et al., 2022). It significantly reduces the risk of inadvertently eliminating important semantic information present in the optical images during the cloud removal process. However, the semantic information in cloudy optical images may be incomplete or distorted, potentially providing misleading information for the learning process. Additionally, the presence of clouds can disrupt the correlation and consistency between optical and SAR images, posing additional challenges for the fusion of these two modalities.

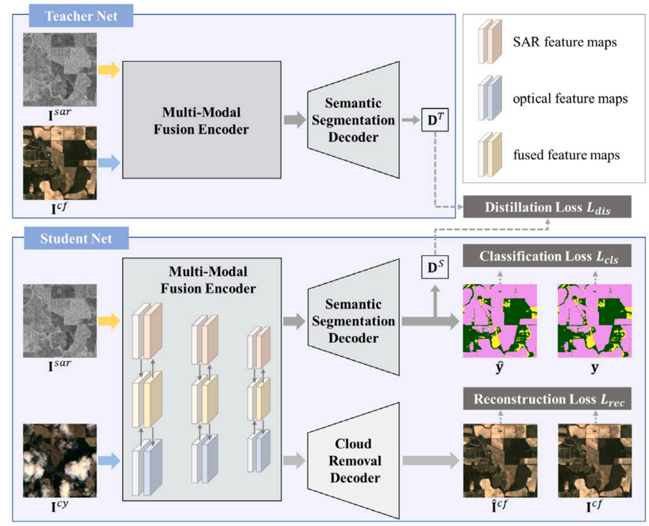

Fig. 3. Overview of the proposed CloudSeg method.

3.2. CloudSeg

3.2.1. Overview

The overall framework of the proposed CloudSeg algorithm is illustrated in Fig. 3. It primarily involves two key parts to mitigate the performance degradation caused by cloud cover: (1) it incorporates a multi-task learning strategy, which includes an additional cloud removal task related to the core land cover classification task, aiming to reduce the semantic uncertainty in cloud-covered areas. (2) It utilizes knowledge distillation, transferring the knowledge learned by the teacher network under cloud-free conditions to the student network, thereby extracting and enhancing critical information from the unobstructed portions of cloud-covered images. Through the synergistic effect of these two key components, CloudSeg can effectively tackle the challenges posed by cloud cover. Details of these two key components are as follows.

Multi-Task Learning. The objective of the land cover classification task is to integrate all potentially valuable information derived from optical and SAR images, ensuring precise pixel-level predictions. To this end, we introduce a cloud removal component to enhance the learning process associated with the classification task. This supplementary step is designed to elevate the information quality of the optical image, facilitating a robust fusion with the SAR image to yield stable outcomes. It is worth noting that the land cover classification task, while enhancing semantic features, substantially advances the model's comprehension of image content, providing more precise guidance for the cloud removal task. This implies that features in both land cover classification and cloud removal tasks are mutually beneficial and exploitable. Therefore, by jointly learning these two tasks, the model can optimize their interdependence during the learning process, ultimately improving the performance of both tasks. Considering the correlation between these two tasks, we adopt a shared encoder to simultaneously handle these two tasks, allowing information to be exchanged and shared between them. In contrast to the sequential approach of initially removing clouds and subsequently classifying land cover, simultaneous processing of both tasks can effectively minimize the potential for error propagation and accumulation.

Despite the tasks of land cover classification and cloud removal exhibiting an intrinsic connection, their optimization goals are distinct, potentially leading to different local optima. To counteract the implicit influence of the auxiliary task on the primary task during the learning process, it is imperative that the output of the auxiliary task operates with a degree of independence from the main task (Ruder, 2017). Therefore, CloudSeg is designed with specialized decoders to individually address land cover classification and cloud removal. Each task is guided by a dedicated loss function to steer its optimization. Specifically, for the land cover classification task, the cross-entropy loss is employed:

$$\mathcal{L}_{cls} = -\frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} \sum_{c=1}^{C} \mathbf{y}_{h,w,c} \log(\hat{\mathbf{y}}_{h,w,c}), \tag{1}$$

where H and W indicate the height and width of the image, respectively. C denotes the number of land cover categories, and $\mathbf{y}_{h,w,c}$ and $\hat{\mathbf{y}}_{h,w,c}$ represent the ground truth label and the predicted probability for a specific pixel located at (h, w) for class c, respectively. For the cloud removal task in CloudSeg, the Charbonnier loss, augmented with extra constraints for cloudy regions, is used (Xu et al., 2024):

$$\mathcal{L}_{rec} = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} (1 + \lambda \mathbf{M}_{h,w}) \times \phi(\hat{\mathbf{I}}^{cf}_{h,w} - \mathbf{I}^{cf}_{h,w}), \tag{2}$$

where $\hat{\mathbf{I}}^{cf}$ and $\mathbf{I}^{cf}$ correspond to the de-clouded and cloud-free images, respectively. ϕ is the Charbonnier loss function. $\mathbf{M}$ represents the cloud mask utilized to determine whether pixels in the image are obscured by clouds. $\mathbf{M}_{h,w} = 1$ indicates that the pixel positioned at (h, w) is obscured by clouds, whereas it equals 0 if the pixel resides in the unobstructed portion. λ indicates the additional constraints applied to cloudy regions.

Knowledge Distillation. Theoretically, even in cases where cloud cover is present, those areas not obscured by clouds retain complete information about the terrain features. Therefore, the classification results for these areas are expected to align with those obtained under cloud-free conditions. However, due to feature interactions between cloud-free and cloudy regions, the presence of clouds can impact the regions of the image that are not directly covered by clouds. To ensure that the information from these parts is undistorted by cloud interference and is accurately represented, we use a teacher model trained on cloud-free data to guide the learning of the model under cloudy conditions, as the teacher model understands the patterns that the data should exhibit in the absence of cloud interference. By transferring this knowledge to the student model, the student model becomes more robust in handling disturbances caused by cloud cover.

Specifically, in scenarios without cloud interference, both optical and SAR images provide a wealth of features crucial for land cover classification. CloudSeg involves training a teacher network capable of effectively integrating cloud-free optical images with SAR images. The teacher network is then used to predict on cloud-free data, generating soft labels, *i.e.*, probability distributions over the land cover types. These soft labels are then employed to train the student network. When the student network makes predictions using data acquired under cloudy conditions, its goal is to mimic the predictive behavior of the teacher network under cloud-free conditions. In other words, the student network's predictions under cloudy conditions should align as closely as possible with those of the teacher network. A straightforward approach would be to align the distributions of the student and teacher networks completely. However, since the information from areas obscured by clouds differs from that under cloud-free conditions, there is a tendency to increase error-prone prediction. Therefore, in the distillation process, we selectively align the distributions only for those areas that are not obscured by clouds. L1 loss is employed to measure the discrepancy between the distributions of the student and teacher models:

$$\mathcal{L}_{dis} = \frac{1}{H \times W} \sum_{h=1}^{H} \sum_{w=1}^{W} (1 - \mathbf{M}_{h,w}) \times |\mathbf{D}^{T}_{h,w} - \mathbf{D}^{S}_{h,w}|, \tag{3}$$

where $\mathbf{D}^T$ represents the soft labels generated by the teacher network when fed with cloud-free optical and SAR images, and $\mathbf{D}^S$ denotes the soft labels produced by the student network when it processes optical images covered with clouds and SAR images.

3.2.2. Network modules

Multi-Modal Fusion Encoder. Considering the correlation between the tasks of land cover classification and cloud removal, we employ a shared encoder, referred to as the multi-modal fusion encoder, to concurrently address these two tasks, allowing information to be exchanged and shared between them. However, each task is optimized to cater to its specific information requirements. The cloud removal task focuses on compensating for information loss in optical images by exploiting the commonalities between optical and SAR images, while the land cover classification task aims to fully exploit both the common and complementary information present in optical and SAR images to achieve a comprehensive representation. Therefore, the multi-modal fusion encoder employs a three-branch network, as illustrated in Fig. 4(a). Specifically, the top and bottom branches are developed for modality-specific representation learning, while the middle branch is designed for modality-shared representation learning. It enables the encoder to not only independently capture the characteristics of each modality but also to integrate these characteristics, thereby better fulfilling the specific requirements of cloud removal and land cover classification tasks.

Specifically, each branch within the multi-modal fusion encoder employs a Segformer encoder (Xie et al., 2021) as its backbone network. Given an optical image $\mathbf{I}^{cy}$ and a SAR image $\mathbf{I}^{sar}$, we first feed them into their respective branches to extract modality-specific features. The modality-shared branch takes an empty tensor as input and hierarchically aggregates the information from modality-specific features. In detail, a Feature Integration Block (FIB) combines F_i^{sar}, F_i^{opt}, and F_i^c to produce fused features denoted as $\tilde{F}_i^c$. Subsequently, a Feature Compensation Block (FCB) utilizes the fused features to enhance the features of individual modalities, resulting in enhanced features $\tilde{F}_i^{sar}$ and $\tilde{F}_i^{opt}$.

The details of the FIB and FCB are illustrated in Fig. 5. To account for potential misalignment between optical and SAR features, a Pyramid Pooling Layer (PPL) is incorporated before feature interaction. Given the feature F_i, PPL captures contextual information P_i for subsequent feature fusion and enhancement. To prioritize features that are less affected by cloud cover, the FIB incorporates a spatial attention mechanism. In cloud-free conditions, optical and SAR features are expected to exhibit relatively consistent descriptions of land cover. However, the presence of cloud cover can lead to an amplification of differences between optical and SAR features. Consequently, the spatial attention weight matrix can be effectively computed by comparing the difference between optical and SAR features. Subsequently, the FIB combines features from different branches to achieve feature fusion, aiming to obtain a more comprehensive feature representation. Considering that the importance of features from various branches may differ, the FIB adaptively learns weight coefficients for each feature channel, highlighting significant features while suppressing less important ones. The fused feature $\tilde{F}_i^c$ is obtained by summing the features from all branches. Finally, the FCB refines the feature of each modality, which computes the difference between the feature of each modality and the fused feature and propagates the information through the use of a gating function.

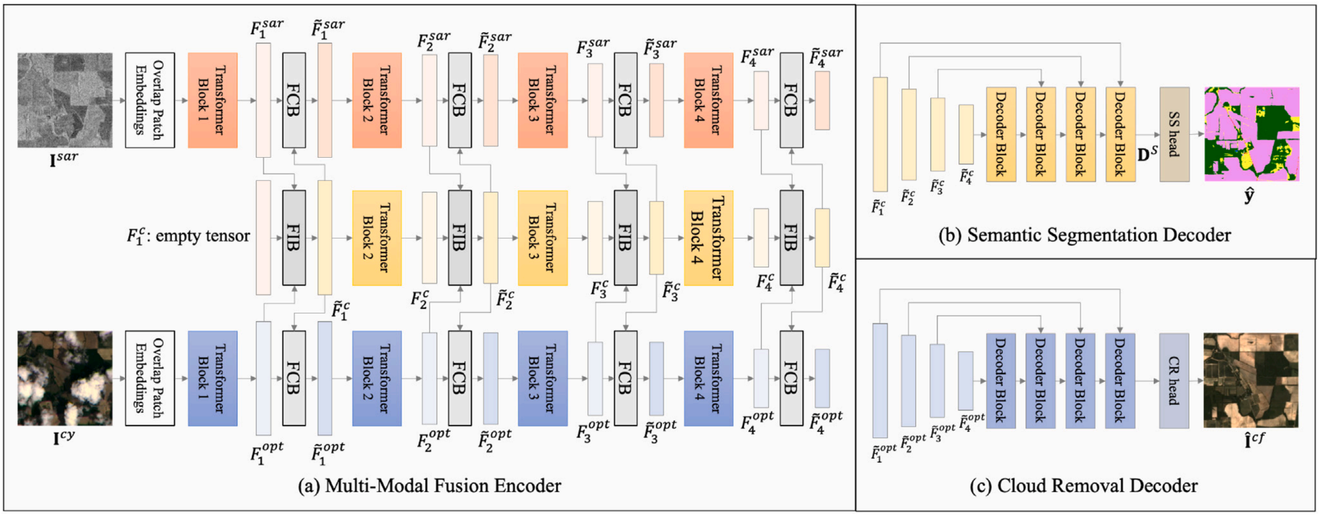

Fig. 4. The network modules of CloudSeg. (a) multi-modal fusion encoder. (b) semantic segmentation decoder. (c) cloud removal decoder.

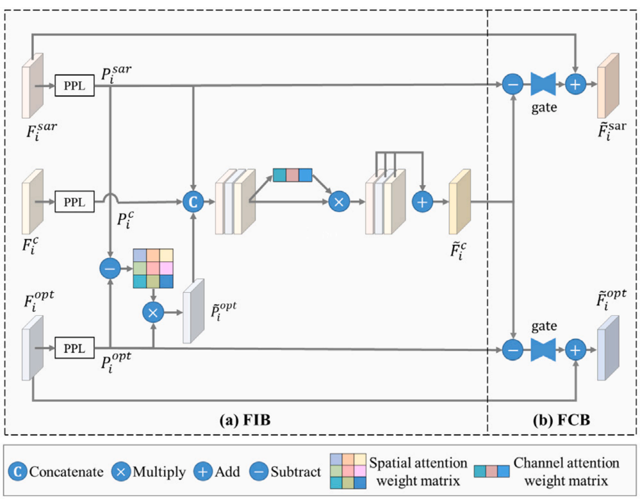

Fig. 5. (a) Detail of the Feature Integration Block (FIB). (b) Detail of the Feature Compensation Block (FCB).

Semantic Segmentation Decoder. The semantic segmentation decoder, as depicted in Fig. 4(b), is based on the U-Net decoder architecture. In the land cover classification task, the utilization of comprehensive features is crucial. Consequently, the semantic segmentation decoder employs fused features $\{\tilde{F}_i^c\}_{i=1}^4$ to distinguish between different land cover categories.

Cloud Removal Decoder. The architecture of the cloud removal decoder closely resembles that of the semantic segmentation decoder, as illustrated in Fig. 4(c). Considering that the objective of the cloud removal task is to reconstruct regions in optical images that are obscured by cloud cover, the cloud removal decoder employs optical features $\{\tilde{F}_i^{opt}\}_{i=1}^4$ for the reconstruction of cloud-free images.

3.2.3. Optimization

The optimization of CloudSeg includes the training of both the teacher network and the student network. The teacher network within the CloudSeg framework leverages cloud-free optical and SAR images to predict the land cover maps. It comprises a multi-modal fusion encoder and a semantic segmentation decoder, which are nearly identical to their counterparts in the student network, with only minor differences. Specifically, the teacher network is tailored to handle images unobstructed by cloud cover, where the optical features it encounters are comprehensive across all regions. Consequently, unlike the student network, it does not integrate the spatial attention mechanism to modify the optical features. During the training phase, the cross-entropy loss is used as the optimization objective.

The objective of the student network is to accurately predict land cover maps in the presence of cloud cover. During its training phase, it incorporates three components: the classification loss $\mathcal{L}_{cls}$, the reconstruction loss $\mathcal{L}_{rec}$, and the distillation loss $\mathcal{L}_{dis}$. the overall function is as follows:

$$\mathcal{L} = \mathcal{L}_{cls} + \alpha * \mathcal{L}_{rec} + \beta * \mathcal{L}_{dis}, \tag{4}$$

where α and β denote the balancing parameters.

4. Experiments

4.1. Experimental settings

Datasets and Metrics. Our experiments are conducted on two benchmark datasets: *M3M-CR* (Xu et al., 2024) and *WHU-OPT-SAR* (Li et al., 2022). The *M3M-CR* dataset features cloud-covered optical images derived from real remote sensing scenarios, while the *WHU-OPT-SAR* dataset does not include cloud-covered images corresponding to its cloud-free counterparts. We perform artificial cloud layer synthesis on the available cloud-free images within the *WHU-OPT-SAR* dataset to simulate the effect of cloud cover. The use of synthetic clouds to evaluate the effectiveness in dealing with cloud cover has been validated in previous studies (Enomoto et al., 2017; Liu et al., 2024). Details of these two datasets are as follows:

- *M3M-CR*. It encompasses a total of 63,000 samples collected from 780 non-overlapping regions of interest (ROIs) across all inhabited continents throughout the meteorological seasons of 2020. These samples are uniformly distributed across various cloud coverage levels. Specifically, 60,000 samples from 660 ROIs are allocated for the training set, with the remaining 3000 samples from 120 ROIs designated for the test set. In our study, we further subdivide the training set by selecting 6494 samples from 65 ROIs as the validation set, with the remaining 53,506 samples from 595 ROIs used for training. Each sample is composed of a quartet of orthorectified, geo-referenced cloudy, and cloud-free optical images, as well as the corresponding SAR image and land cover map. The optical images are sourced from the PlanetScope Level-3B top-of-atmosphere reflectance product, which include four spectral bands (RGB and NIR) with a spatial resolution of 3 m and a size of 300 × 300 pixels. The SAR

Table 1
Quantitative comparisons of proposed CloudSeg networks to state-of-the-art methods, with the best-performing results indicated in bold and the next-best results underlined. CloudSeg* is a variant of CloudSeg, which is trained solely based on classification loss for the semantic segmentation task without employing multi-task learning or knowledge distillation strategies.

	M3M-CR						*WHU-OPT-SAR*					
	Cloud-Free		Cloudy		Overall		Cloud-Free		Cloudy		Overall	
	mIoU	mPA	mIoU	mPA	mIoU	mPA	mIoU	mPA	mIoU	mPA	mIoU	mPA
HAFNetE (Ferrari et al., 2021)	47.10	61.66	44.85	58.20	46.22	60.21	42.73	52.81	39.17	49.03	40.90	50.86
DCSA-Net (Wang et al., 2022)	39.32	52.02	37.33	49.73	38.49	51.07	31.73	40.81	30.35	38.72	31.06	39.74
MCANet (Li et al., 2022)	49.19	62.88	46.20	58.80	47.96	61.17	48.93	60.47	44.86	55.63	46.84	57.93
AMM-FuseNet (Ma et al., 2022)	49.18	62.80	46.34	58.41	48.02	60.90	47.26	58.21	43.56	54.06	45.35	56.04
PAGNet (Liu et al., 2022)	43.85	55.58	43.11	53.92	43.72	55.02	35.84	45.32	34.34	43.42	35.09	44.34
CMX (Zhang et al., 2023)	51.61	65.02	48.80	61.00	50.48	63.34	46.19	56.74	44.40	54.48	45.27	55.57
CloudSeg* (Ours)	52.92	66.79	49.98	62.56	51.72	64.98	50.84	61.83	48.85	59.73	49.82	60.75
CloudSeg (Ours)	**55.91**	**69.02**	**50.87**	**63.43**	**53.64**	**66.57**	**53.53**	**65.67**	**51.36**	**63.34**	**52.40**	**64.47**

images originate from the Sentinel-1 Level-1 GRD product with a spatial resolution of 10 m, which are acquired in IW mode with two polarization channels (VV and VH). The size of SAR images is 90 × 90 pixels. Following previous practices (Xu et al., 2024), we upsample the SAR images to match the size of the optical images using nearest neighbor interpolation.

- *WHU-OPT-SAR*. It comprises 100 distinct ROIs located in Hubei Province, China, where optical images (four channels in RGB and near-infrared) and SAR images are derived from GF-1 and GF-3, respectively, with a spatial resolution of 5 m. We allocate the dataset as follows: 60 ROIs for training, 20 ROIs for validation, and 20 ROIs for testing. With a sliding window of 256 × 256 pixels for cropping, we ultimately derive 9922 training samples, 2537 validation samples, and 3042 test samples. To simulate cloud interference, Perlin noise is evenly added to each band of the optical images. Specifically, the training, validation, and test datasets are each uniformly divided into five subsets to simulate various cloud coverage levels. These subsets contain optical images with simulated cloud coverage ranging from 0% to 20%, 20% to 40%, 40% to 60%, 60% to 80%, and 80% to 100%, respectively.

The performance of land cover classification is evaluated using mean Intersection over Union (mIoU) and mean Pixel Accuracy (mPA). mIoU quantifies the overlap between predicted land cover maps and ground truth by calculating the average ratio of their intersection to their union across different classes. It primarily assesses the segmentation precision and consistency of the model. Meanwhile, mPA computes the proportion of correctly classified pixels to the total number of pixels, averaged across all classes, offering insight into the model's accuracy at the pixel level. The formulas of mIoU and mPA are expressed as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FP_c + FN_c}, \tag{5}$$

$$\text{mPA} = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FP_c}, \tag{6}$$

where C denotes the total number of land cover classes, TP_c denotes the number of true positive pixels for class c, FP_c denotes the number of false positive pixels for class c, and FN_c denotes the number of false negative pixels for class c.

Implementation Details. Our network is implemented using Pytorch on a single NVIDIA Geforce RTX 3090 GPU. The Adam optimizer is used, with a learning rate of 10^{-4} that decays by 50% every 5 epochs. The batch size is set to 32 and the maximum epoch of training iterations is set to 30. λ within $\mathcal{L}_{rec}$ is set to 5 following the configuration in Xu et al. (2024), and the weighting factors α and β are both set to 1 in an empirical manner.

4.2. Comparisons with state-of-the-art methods

We compare the proposed CloudSeg networks to state-of-the-art multi-modal semantic segmentation methods, including HAFNetE (Ferrari et al., 2021), DCSA-Net (Wang et al., 2022), MCANet (Li et al., 2022), AMM-FuseNet (Ma et al., 2022), PAGNet (Liu et al., 2022), and CMX (Zhang et al., 2023). To validate the superiority of CloudSeg in mitigating the performance degradation caused by cloud cover, we also train a variant of CloudSeg, denoted as CloudSeg*, solely based on classification loss for the semantic segmentation task, *i.e.*, without employing multi-task learning or knowledge distillation strategies. The quantitative results are presented in Table 1. To provide a more comprehensive analysis, our evaluation not only focuses on overall performance but also separately assesses the classification results in both cloud-covered and cloud-free regions. The proposed CloudSeg networks' performance is notably superior compared to existing state-of-the-art methods across both datasets, whether in cloud-covered or cloud-free regions. Specifically, without employing multi-task learning and knowledge distillation strategies, our method achieves an mIoU of 51.72% on the *M3M-CR* dataset and 49.82% on the *WHU-OPT-SAR* dataset, surpassing existing approaches. It demonstrates the ability of the proposed network to effectively integrate valuable information from SAR images and cloud-covered optical images. Moreover, the inclusion of multi-task learning and knowledge distillation strategies further elevates our method's performance. By reducing semantic ambiguity in cloudy areas and enhancing effective information in cloud-free areas, our method can better address the negative impacts of cloud cover and outperform state-of-the-art multi-modal semantic segmentation methods with a gain of about 3.16% in terms of mIOU on the *M3M-CR* dataset and 5.56% on the *WHU-OPT-SAR* dataset.

We visualize the results of the land cover classification in Figs. 6 and 7, where we showcase two typical scenes from both the *M3M-CR* dataset and the *WHU-OPT-SAR* dataset for qualitative evaluation. The results obtained by CloudSeg exhibit a higher level of consistency with the ground truth when compared to other methods. While CloudSeg* shows segmentation contours similar to those of CloudSeg, it lags in terms of classification accuracy. For instance, in the first scene of Fig. 6, the water bodies are challenging to identify in optical images but are distinctly recognizable in SAR images. CloudSeg* and all other competing methods, fusing these two modalities of images, fail to provide accurate predictions. Additionally, in the same scene, the barren land areas, while clearly visible in both optical and SAR images, still exhibit certain predictive discrepancies in CloudSeg* and all other competing methods. It highlights the intricate impact of cloud cover on multi-modal fusion. The proposed CloudSeg, tailored to reduce semantic ambiguity in cloud-covered areas and enhance the effective information in cloud-free regions, successfully counteracts the adverse effects of cloud interference and attains more accurate identification of water bodies and barren land. Furthermore, in the second scene depicted in Fig. 7, CloudSeg presents a continuous segmentation result for roads, whereas other methods exhibit fragmented or discontinuous

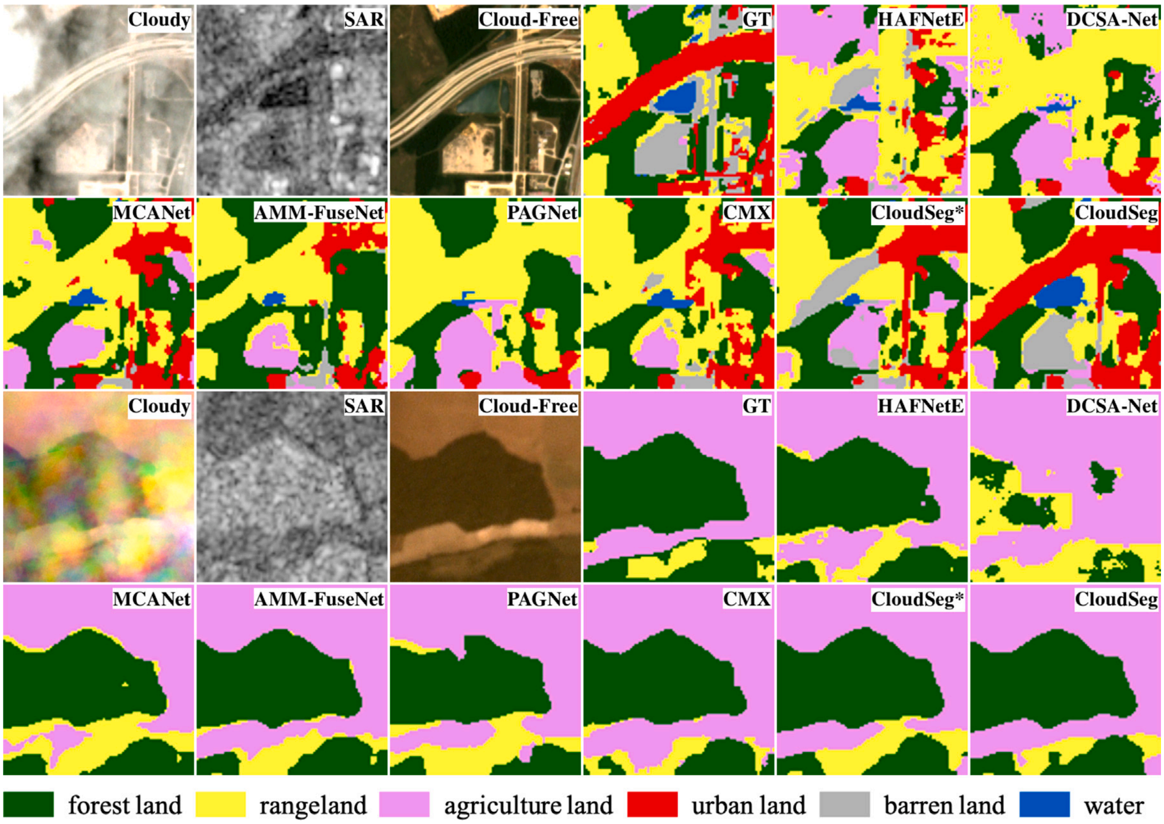

Fig. 6. Visualization of semantic segmentation results for two scenes in the *M3M-CR* dataset.

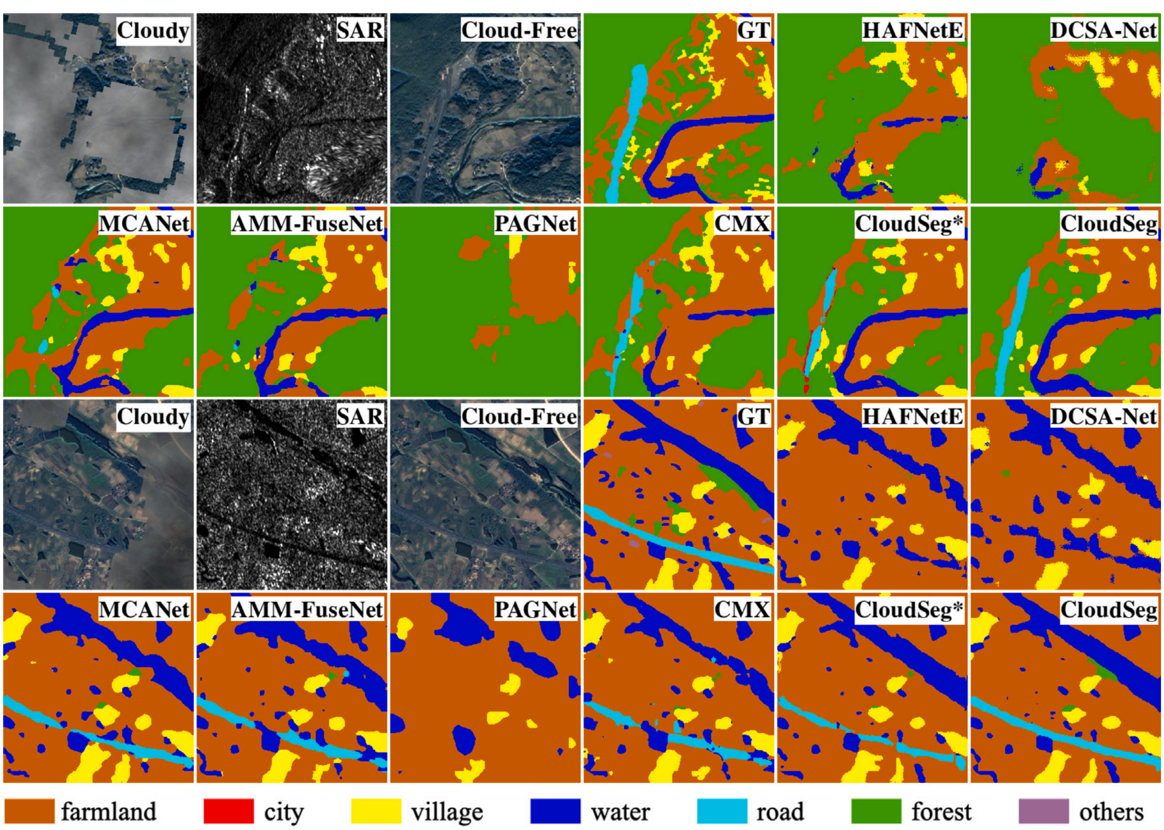

Fig. 7. Visualization of semantic segmentation results for two scenes in the *WHU-OPT-SAR* dataset.

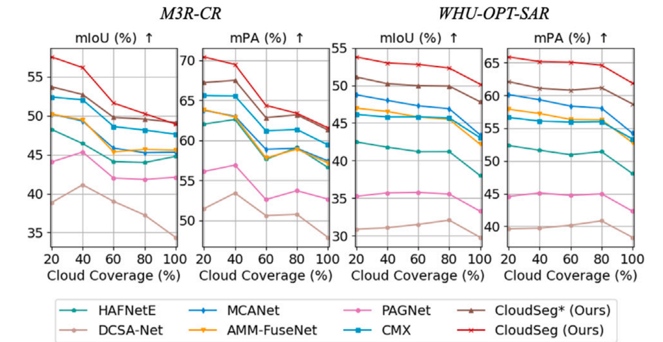

Fig. 8. Quantitative comparisons of proposed CloudSeg networks to state-of-the-art methods on different cloud cover levels.

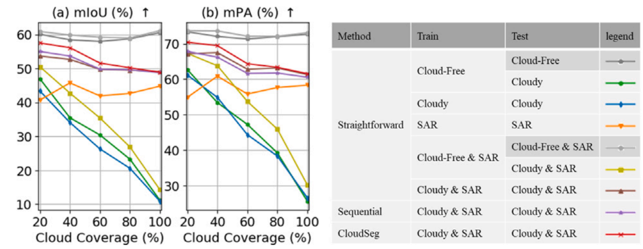

Fig. 9. Quantitative comparisons of CloudSeg to naive strategies on the *M3M-CR* dataset over different cloud cover levels.

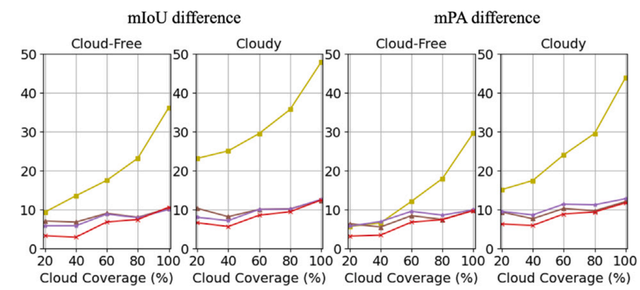

Fig. 10. The performance degradation experienced by different models due to cloud coverage in the context of SAR-optical data fusion. The legend is consistent with that of Fig. 9.

patterns. It further demonstrates the superiority of CloudSeg in semantic segmentation of land cover under cloudy conditions. It is noteworthy that, although CloudSeg exhibits a notable qualitative enhancement in segmentation results compared to CloudSeg*, this enhancement may only involve a small portion of the overall pixels. Consequently, it leads to subtle disparities in metrics like mIoU and mAP, making the improvements less observable in quantitative terms.

We further assess the classification performance on different cloud cover levels, as shown in Fig. 8. The proposed methods perform favorably when compared with state-of-the-art methods across all levels of cloud cover, showing a distinct negative correlation between the accuracy of classification and the degree of cloud coverage. Additionally, we observe that the performance improvement of our method is more pronounced in the *WHU-OPT-SAR* dataset than in the *M3M-CR* dataset. This discrepancy is primarily attributed to the fact that the cloud-covered optical images in the *WHU-OPT-SAR* dataset are synthetic, allowing precise identification of areas affected by cloud coverage. In contrast, the cloud-covered optical images in the *M3M-CR* dataset originate from real remote sensing scenarios, and the associated cloud masks, sourced from Planet Labs, exhibit a somewhat lower level of precision. The loss functions for multi-task learning and knowledge distillation rely on these cloud masks, which contributes to the variance in performance improvement observed between the two datasets. It also suggests that with more accurate cloud masks, the proposed method has the potential to achieve a better performance. CloudSeg reduces semantic ambiguity in areas of the cloudy image that are obscured by clouds and enhancing effective information in the unobstructed portion

4.3. Comparisons with naive strategies

To further validate the superiority of CloudSeg in mitigating the interference of cloud cover on land cover classification, we compare CloudSeg to the naive strategies, *i.e.*, the straightforward and sequential paradigm. The results are shown in Table 2. For the straightforward paradigm, except for considering the combination of cloud-covered optical and SAR images as the training set, we also consider four other different training sets: cloud-free optical images, cloud-covered optical images, SAR images, and a combination of cloud-free optical and SAR images. During the testing phase, when dealing with cloudy conditions, only cloudy optical and SAR images are available. However, in cases where the optical images within the training set are cloud-free, we also evaluate the trained model's performance under cloud-free conditions as a baseline, with results shown in gray. For the sequential paradigm, we employ the well-trained Align-CR model in Xu et al. (2024) to remove clouds from cloud-covered optical images and subsequently combine them with SAR images for the land cover classification task. Additionally, Fig. 9 shows the overall performance of each model under different levels of cloud coverage.

We can find that under cloud-free conditions, the accuracy of land cover classification is significantly higher compared to cloudy conditions. Furthermore, the incorporation of SAR images for fusion leads to further accuracy improvements. This enhancement can be attributed to the additional information provided by SAR images in conjunction with optical images.

In unavoidable cloudy situations, our framework, which addresses the negative impacts of cloud cover by reducing semantic ambiguity in cloud-covered areas and enhancing the effective information in cloud-free regions, achieves the best performance. When predicting land cover maps directly from data acquired under cloudy conditions, we can observe that models trained with cloud-free images suffer a significant decrease in performance when presented with cloudy inputs. While the models are trained with cloudy images, the performance degradation is significantly mitigated. Since the models are more robust to the distribution of cloudy images presented in the training data. Additionally, we conduct separate evaluations of performance degradation in both cloud-covered and cloud-free areas. In the context of SAR-optical data fusion, the performance of the model trained with cloud-free optical and SAR images, when using data acquired under cloud-free conditions as input, can serve as a benchmark for the optimal accuracy that could potentially be achieved under cloudy conditions when using the same network architecture. By comparing this benchmark with the performance of models that use data acquired under cloudy conditions as input, we can evaluate the performance degradation experienced by each model due to cloud coverage, as shown in Fig. 10. We can observe that performance degradation occurs not only in cloud-covered regions but also in cloud-free regions, *i.e.*, the disruption caused by clouds on land cover classification not only affects the directly covered regions but also the adjacent cloud-free regions. As the level of cloud cover increases, the performance degradation in both regions worsens. Specifically, cloud-covered regions exhibit a more severe performance degradation compared to cloud-free regions, due to the reduction of optical information in those regions.

SAR images perform relatively stable in terms of predicting land cover maps over different cloud cover levels, which perform worse than cloud-free optical images but better than cloudy optical images affected by significant cloud interference. When integrating optical and

Table 2
Quantitative comparisons of CloudSeg to naive strategies on the *M3M-CR* dataset. The 'Train' column specifies the datasets used for training the model, while the 'Test' column specifies the conditions under which the trained model is evaluated. The results obtained under cloud-free conditions are shown in gray. Note: in the 'Train' and 'Test' columns, 'Cloudy' refers to cloud-covered images, while 'Cloud-Free' denotes the corresponding images without cloud cover. In the table header, 'Cloudy' and 'Cloud-Free' indicate the regions within the cloud-covered images that are obscured and unobscured, respectively. When evaluating the model's performance under cloud-free conditions, we still consider these specific regions to maintain consistency in the evaluation process.

Method	Train	Test	Cloud-Free		Cloudy		Overall	
			mIoU	mPA	mIoU	mPA	mIoU	mPA
Straightforward	Cloud-Free	Cloud-Free	59.63	72.94	59.73	72.12	59.83	72.64
		Cloudy	40.89	57.70	19.94	35.33	30.57	46.53
	Cloudy	Cloudy	50.79	64.07	36.70	48.35	43.88	56.58
	SAR	SAR	42.35	56.82	44.46	58.53	43.55	57.82
	Cloud-Free & SAR	Cloud-Free & SAR	60.37	73.41	60.61	72.79	60.63	73.21
		Cloudy & SAR	46.05	64.10	23.71	41.60	34.94	52.80
	Cloudy & SAR	Cloudy & SAR	52.92	66.79	49.98	62.56	51.72	64.98
Sequential	Cloudy & SAR	Cloudy & SAR	53.69	66.48	50.08	61.63	52.15	64.35
CloudSeg	Cloudy & SAR	Cloudy & SAR	**55.91**	**69.02**	**50.87**	**63.43**	**53.64**	**66.57**

Table 3
Quantitative ablation study of multi-task learning and knowledge distillation strategies on the *M3M-CR* dataset.

w/ CR	w/ KD	Cloud-Free		Cloudy		Overall	
		mIoU	mPA	mIoU	mPA	mIoU	mPA
–	–	52.92	66.79	49.98	62.56	51.72	64.98
✓	–	54.61	67.92	50.54	62.92	52.83	65.73
–	✓	54.71	68.66	50.66	**63.64**	52.94	66.45
✓	✓	**55.91**	**69.02**	**50.87**	63.43	**53.64**	**66.57**

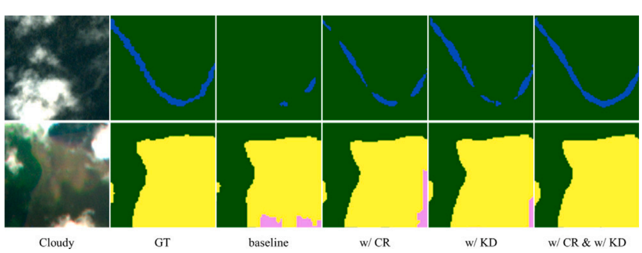

Fig. 11. Qualitative ablation study of multi-task learning and knowledge distillation strategies on the *M3M-CR* dataset. The legend is consistent with that of Fig. 6.

SAR images, the performance improvement is more significant under cloudy conditions compared to cloud-free conditions, in comparison to using only optical images. This is because SAR images not only offer unique insights that are not available in the cloud-free regions of optical images but also effectively compensate for the missing information in the cloud-covered regions of these optical images.

We further compare the sequential paradigm to the straightforward approach and observe that the former exhibits relatively superior performance in terms of mIoU but slightly lags in mPA. This is attributed to the cloud removal process, employed as a preprocessing step, which effectively reconstructs the structural and morphological features of cloud-covered areas, thereby leading to an improved mIoU. However, at a finer level of detail, such as texture information, the pixel-level accuracy may be compromised due to blurring effects or minor errors that occur during the process.

Instead of directly performing the prediction, CloudSeg simultaneously predicts land cover maps and reconstructs cloud-free images. Through the cooperation between the low-level image restoration and the high-level semantic understanding tasks, CloudSeg effectively reduces the ambiguity caused by the clouds. Furthermore, CloudSeg leverages knowledge distillation to enhance effective information in cloud-free regions. In both cloud-obscured and cloud-free areas, CloudSeg exhibits superior performance compared to other methods.

4.4. Ablation study

CloudSeg primarily involves two key components—multi-task learning and knowledge distillation—to effectively counteract the performance degradation attributed to cloud cover. To evaluate the impact of each component, we compare several variants with and without the auxiliary cloud removal (CR) task and knowledge distillation (KD). The qualitative and qualitative results are shown in Table 3 and Fig. 11. We find that employing either the multi-task learning strategy or the knowledge distillation strategy alone results in a certain degree of performance improvement. However, the combination of these two strategies leads to a more significant enhancement in performance. As illustrated in the first scene depicted in Fig. 11, when neither strategy is used, water is barely distinguished, whether in cloud-covered or cloud-free areas. When employing either strategy alone, the classification accuracy of water improves, but the improvement is localized differently. For instance, employing only the multi-task learning strategy yields better results at the bottom of the image where water is obscured by clouds, while employing only the knowledge distillation strategy improves results in the upper left corner of the image, where the water is not obscured by cloud cover. However, employing these strategies separately still results in certain discontinuities. By integrating both strategies, water is segmented both accurately and completely. This is primarily due to the intrinsic connection and mutual dependence between the features of cloud-covered and cloud-free areas. The introduction of the cloud removal task reduces the semantic ambiguity in cloud-covered areas, thereby diminishing their interference in cloud-free regions. Moreover, the incorporation of a knowledge distillation strategy enhances effective information in cloud-free areas, facilitating a more accurate restoration process in cloud-covered areas. Because the restoration of cloud-covered areas relies not only on SAR images but also on the contextual information from adjacent cloud-free areas (Xu et al., 2022, 2024). In turn, the optimization of the cloud removal task further minimizes interference in cloud-free areas. Therefore, by adopting both of these core strategies simultaneously, CloudSeg achieves maximal performance improvement.

CloudSeg transfers the knowledge learned by the teacher network under cloud-free conditions to the student network, aiding the student network in overcoming cloud interference and accurately representing the features of the unobstructed portions within cloud-covered images. Specifically, CloudSeg aligns features of the unobstructed portions within cloud-covered images to avoid the performance degradation that could arise from misaligning information in areas obscured by clouds. To validate the effectiveness of our approach, we also experiment with aligning features across all regions within the images, regardless of whether these areas are affected by clouds. As shown in Table 4, we

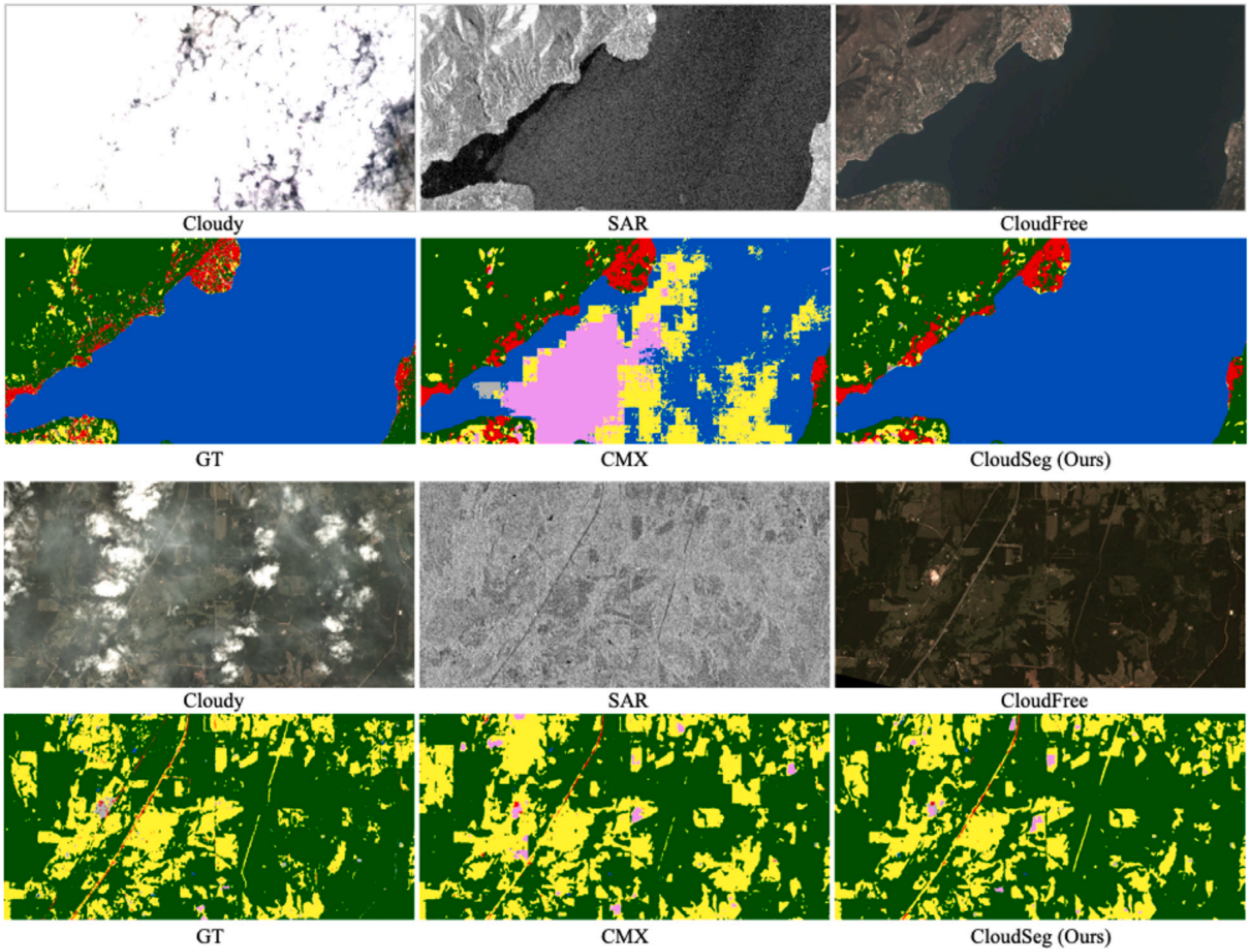

Fig. 12. The performance of CloudSeg for large-area land cover mapping. The legend is consistent with that of Fig. 6.

Table 4
Comparative evaluation of region-specific alignment in knowledge distillation on the *M3M-CR* dataset.

Distillation area	Cloud-Free		Cloudy		Overall	
	mIoU	mPA	mIoU	mPA	mIoU	mPA
None	54.61	67.92	50.54	62.92	52.83	65.73
All	54.87	68.64	50.36	63.08	52.87	66.18
Cloud-Free	**55.91**	**69.02**	**50.87**	**63.43**	**53.64**	**66.57**

Table 5
Quantitative comparisons of CloudSeg's cloud removal component to state-of-the-art cloud removal methods.

Method	PSNR ↑ (dB)	SSIM ↑	SAM ↓ (°)	MAE ↓ (10^{-2})
DSen2-CR	28.86	0.882	6.68	2.93
GLF-CR	30.00	0.906	5.65	2.55
Align-CR	30.54	0.914	5.35	2.37
CloudSeg (Ours)	**31.06**	**0.917**	**5.27**	**2.25**

observe that forcing feature alignment across all regions results in a drop in classification performance in both cloud-covered and cloud-free areas. In comparison to the model without feature alignment, it shows a minor improvement in cloud-free regions but a decline in cloud-covered areas. Due to the mismatched information between the cloud-covered portions of optical images and their corresponding parts in cloud-free optical images, attempting to align these mismatched data may lead to overfitting in the student model, resulting in a decline in its predictive performance in cloud-covered areas.

5. Discussion

CloudSeg for Large-Area Land Cover Mapping. We further apply our CloudSeg to two large areas using Planetscope optical data and Sentinel-1 SAR data, and compare it with the top-performing CMX method, as illustrated in Fig. 12. It can be observed that despite CMX being trained on cloudy data to adapt to the presence of clouds, its results are still disrupted by cloud interference when fusing cloudy optical images with SAR images, leading to inaccurate predictions. In contrast, our CloudSeg, designed to account for cloud interference, produces land cover maps that highly align with the ground truth.

Side Product of CloudSeg. CloudSeg integrates an auxiliary cloud removal task to enhance the learning process of its core land cover classification task. Consequently, CloudSeg not only predicts land cover maps but also simultaneously generates cloud-free optical images. In this study, we compare the cloud removal component of CloudSeg with state-of-the-art cloud removal methods, including DSen2-CR (Meraner et al., 2020), GLF-CR (Xu et al., 2022), and Align-CR (Xu et al., 2024), as shown in Table 5. It can be observed that the side product of Cloudseg achieves the best performance, indicating that the task of land cover classification can synergistically enhance the effectiveness of the cloud removal task.

Adaptability of CloudSeg. CloudSeg, built upon the SegFormer architecture, inherits its fundamental characteristics, such as memory consumption, training efficiency, and inference speed. Notably, CloudSeg offers flexibility for integration with other semantic segmentation architectures, enabling customization and optimization for specific tasks and scenarios. For instance, the integration of a lightweight network can enhance operational efficiency, resulting in faster processing speeds and reduced computational costs. The incorporation of a fine-grained feature extraction module can enhance CloudSeg's ability to process high-resolution remote sensing images, highlighting its versatility across diverse applications.

Limitation of CloudSeg. In CloudSeg, we strategically align the predictions of the student and teacher networks only in cloud-free regions. This deliberate approach avoids performance degradation associated with the forced alignment of predictions in cloudy areas, where the information does not match that in cloud-free conditions, as discussed in Section 4.4. It also indicates the significance of precise cloud mask for CloudSeg's overall effectiveness, as mentioned in Section 4.2. Despite cloud detection being a long-studied field, due to the variability and complexity of cloud patterns and atmospheric conditions, it inherently entails the issue of error accumulation when used as a preprocessing step. This challenge inspires us to incorporate an additional cloud detection task within CloudSeg's multi-task framework, aiming further to improve accuracy in land cover classification under cloudy conditions.

Furthermore, the alignment of predictions between the student and teacher networks is based on the assumption that there are no land cover changes between cloud-obscured and corresponding cloud-free images, as forcing alignment in regions with mismatched information can lead to performance degradation, as previously discussed. However, capturing both cloudy and cloud-free views of a specific location simultaneously is impractical. Consequently, the cloud-free observations used as reference are typically temporally close to the cloudy observations, resulting in some inevitable land cover changes. In our study, we conduct land cover mapping under real-world cloudy conditions using the M3M-3R dataset, which is specifically curated to minimize land cover changes. The paired cloudy and cloud-free images are collected with a very short temporal offset, averaging just 2.8 days between them, making the land cover changes generally negligible. If the interval between acquiring cloud-covered and cloud-free images is not strictly controlled during the selection of training data, the impact of land cover changes has to be accounted for. In such cases, it is advisable to first perform change detection on the unobscured areas of the cloudy image, and then align only those regions that exhibit no changes.

Future Enhancement of CloudSeg. Satellite sensors can continuously observe the same region on Earth, thereby obtaining multi-temporal and multi-modal remote sensing images. These images can compensate for the limitation of a single image source, enriching the information completeness. In this study, considering both reliability and timeliness, we select SAR images that are close in time interval to the optical images as auxiliary information. Additionally, the dynamic nature of clouds means that data obtained at different times can provide clear surface information for areas that are obscured by clouds. In future research, we will integrate the fusion of multi-temporal and multi-modal data to advance the accuracy of land cover mapping, in which we will tackle the more complex issues associated with cloud interference.

6. Conclusion

We have presented CloudSeg, a novel multi-modal learning framework designed for semantic segmentation of land cover in cloudy conditions. A key aspect of our approach involves incorporating a low-level visual task, cloud removal, to counteract the adverse effects of semantic ambiguity induced by cloud cover. Moreover, CloudSeg leverages knowledge transfer from cloud-free conditions to enhance multi-modal features within cloud-covered scenarios, optimizing unobstructed regions in cloudy images. The synergistic integration of these two components allows CloudSeg to effectively address the challenges associated with cloud cover in land cover classification. Experimental results on *M3R-CR* and *WHU-OPT-SAR* datasets demonstrate the effectiveness of our approach in improving the performance of land cover classification across all levels of cloud cover.

CRediT authorship contribution statement

Fang Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Data curation. **Yilei Shi:** Writing – review & editing, Methodology. **Wen Yang:** Writing – review & editing, Supervision, Funding acquisition. **Gui-Song Xia:** Writing – review & editing, Supervision. **Xiao Xiang Zhu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research was supported in part by the National Natural Science Foundation of China (NSFC) Regional Innovation and Development Joint Fund under Grant U22A2010 and in part by the NSFC General Program, China under Grant 62271355. The work of X. Zhu is supported by the Munich Center for Machine Learning.

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